

# MODULE 4 - ACTIVITY NO. 2

|                  |                       |            |                   |
|------------------|-----------------------|------------|-------------------|
| Name:            | Buencamino, Ezrael A. | Date:      | 09/05/2025        |
| Subject/Section: | Discrete Math         | Professor: | Rutaqio, Larry T. |

NO. \_\_\_\_\_  
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1. Out of 7 consonant and 4 vowels, how many words of 3 consonant and 2 vowels can be formed?

$$= \binom{7}{3} \cdot \binom{4}{2} \cdot 5!$$

$$= 35 \cdot 6 \cdot 120$$

$$= 25,200$$

2. Number of arrangement of the letters of mathematics so that vowels always come together.

$$= \frac{8!}{2!1!} \times \frac{4!}{2!}$$

$$= \frac{40320}{4} \times \frac{24}{2}$$

$$= 10080 \times 12$$

$$= 120,960$$

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3. How many 3 letter words with or without meaning, can be formed out of the letters of the word 'LOGARITHM', if repetition of letters is not allowed.

$$= P(10, 3)$$

$$= 10 \times 9 \times 8$$

$$= 720$$

4. In how many ways can the letters of the word 'LOGARITHM' be arranged so that the letters 'L' and 'O' are always together?

$$= 3! \times 3!$$

$$= 6 \times 6$$

$$= 36$$